

Completed Pandrosion Prime Atlases and Phase Destruction

The archimedean chart, phase velocity, and zero candidates

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May 2026

Abstract

Paper 025 introduced the prime Pandrosion atlas: each Euler factor is an infinite geometric cell

$$(1 - \ell^{-s})^{-1} = 1 + \ell^{-s} + \ell^{-2s} + \dots,$$

and the critical line $\Re(s) = 1/2$ is the equilibrium locus between the direct chart ℓ^{-s} and the reciprocal chart $\ell^{-(1-s)}$. The present paper adds the missing archimedean chart. The completed zeta factor

$$C(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$$

is treated as the infinite-place Pandrosion chart, while finite Euler cells

$$E_{\ell,N}(s) = 1 + \ell^{-s} + \dots + \ell^{-(N-1)s}$$

form the prime charts. The completed finite atlas is

$$\mathcal{X}_{\mathcal{P},N}(s) = C(s) \prod_{\ell \in \mathcal{P}} E_{\ell,N}(s).$$

The main theorem is a phase-velocity identity. Along any vertical line $s = \sigma + it$, away from zeros and poles,

$$\frac{d}{dt} \arg \mathcal{X}_{\mathcal{P},N}(\sigma + it) = \operatorname{Re} \left[\frac{C'}{C}(s) + \sum_{\ell \in \mathcal{P}} \frac{E'_{\ell,N}}{E_{\ell,N}}(s) \right].$$

On the critical line, the archimedean part is the Riemann–Siegel theta velocity

$$\vartheta'(t) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi.$$

Thus the completed Pandrosion atlas separates the zero problem into two explicit pieces: archimedean phase drift and prime-cell phase interference. We then formulate the phase-destruction conjecture: nontrivial zeros of $\xi(s)$ should arise as limiting points where the balanced completed atlas has simultaneous phase alignment and amplitude collapse. This is not a proof of the Riemann hypothesis; it is a concrete Pandrosion phase calculus extending the chart-equilibrium theorem of Paper 025.

One-sentence summary. Paper 025 finds the critical line from direct/reciprocal chart balance; Paper 026 adds the gamma chart and studies the phase velocity whose destructive alignment should produce zeta zeros.

Status. The finite phase-velocity identities are proved. The convergence of finite-atlas phase-destruction events to nontrivial zeros is stated as a conjecture and a numerical research program.

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1 What 026 adds

Paper 025 gave a geometric explanation for the critical line. It did not study zero locations. The reason is structural: a line is selected by logarithmic chart balance, but zeros require phase cancellation. The zeta function also has an archimedean factor. Ignoring it loses the Riemann–Siegel theta term, which controls the deterministic part of the phase on the critical line.

The present paper therefore adds one object:

$$C(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

The completed zeta function is

$$\xi(s) = C(s)\zeta(s), \quad \xi(s) = \xi(1-s).$$

In the Pandrosion atlas language, $C(s)$ is the archimedean chart, while the Euler factors are the prime charts.

2 Finite completed Pandrosion atlas

Let $\mathcal{P}$ be a finite set of primes and $N \geq 2$. Define

$$E_{\ell,N}(s) = \sum_{r=0}^{N-1} \ell^{-rs} = \sum_{r=0}^{N-1} q_{\ell}(s)^r, \quad q_{\ell}(s) = \ell^{-s}.$$

The finite Euler-Pandrosion product is

$$\mathcal{Z}_{\mathcal{P},N}(s) = \prod_{\ell \in \mathcal{P}} E_{\ell,N}(s),$$

and the completed finite atlas is

$$\mathcal{X}_{\mathcal{P},N}(s) = C(s)\mathcal{Z}_{\mathcal{P},N}(s).$$

Remark 2.1. *For fixed $\mathcal{P}, N$, this is not the completed zeta function. It is a finite model that exposes the two pieces of the phase: $C(s)$ from the infinite place and $E_{\ell,N}$ from the prime geometric cells.*

Completed Pandrosion prime atlas: archimedean chart + prime geometric cells

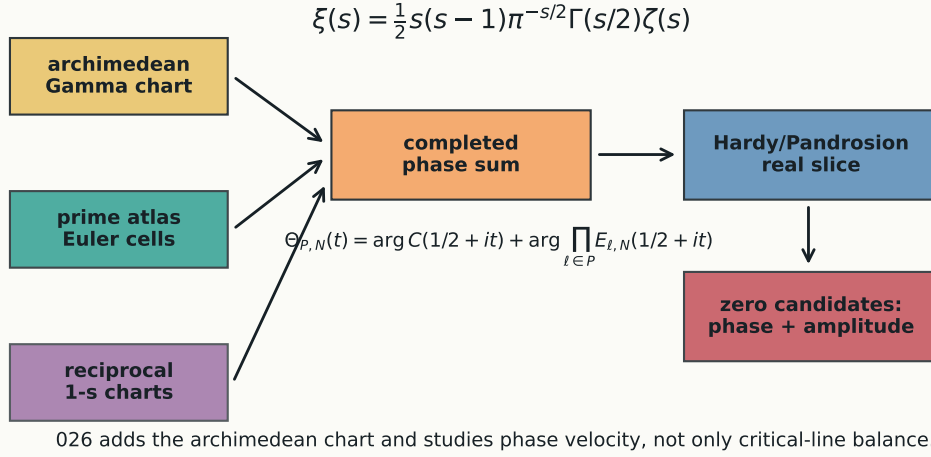


Figure 1: Completed Pandrosion prime atlas. The new ingredient beyond Paper 025 is the archimedean gamma chart $C(s)$. Zero candidates require both phase alignment and amplitude collapse.

3 Phase velocity

Definition 3.1 (Vertical phase velocity). *Let F be holomorphic and nonzero near $s = \sigma + it$. Its vertical phase velocity is*

$$V_F(\sigma, t) = \frac{d}{dt} \arg F(\sigma + it).$$

Lemma 3.2 (Basic phase derivative). *If $F(s) \neq 0$, then*

$$V_F(\sigma, t) = \operatorname{Re} \frac{F'}{F}(\sigma + it).$$

Proof. Along the vertical path $s(t) = \sigma + it$,

$$\frac{d}{dt} \log F(s(t)) = i \frac{F'}{F}(s(t)).$$

Taking imaginary parts gives

$$\frac{d}{dt} \arg F(s(t)) = \operatorname{Im} \left(i \frac{F'}{F}(s(t)) \right) = \operatorname{Re} \frac{F'}{F}(s(t)).$$

□

Theorem 3.3 (Completed finite-atlas phase velocity). *Away from zeros and poles of $\mathcal{X}_{P,N}$,*

$$\frac{d}{dt} \arg \mathcal{X}_{P,N}(\sigma + it) = \operatorname{Re} \left[\frac{C'}{C}(s) + \sum_{\ell \in P} \frac{E'_{\ell,N}}{E_{\ell,N}}(s) \right]_{s=\sigma+it},$$

where

$$\frac{C'}{C}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2),$$

and

$$E'_{\ell,N}(s) = -(\log \ell) \sum_{r=1}^{N-1} r \ell^{-rs}.$$

Proof. The logarithmic derivative of a product is the sum of logarithmic derivatives:

$$\frac{\mathcal{X}'_{\mathcal{P},N}}{\mathcal{X}_{\mathcal{P},N}} = \frac{C'}{C} + \sum_{\ell \in \mathcal{P}} \frac{E'_{\ell,N}}{E_{\ell,N}}.$$

The displayed formula for C'/C follows by differentiating

$$\log C(s) = \log(1/2) + \log s + \log(s-1) - \frac{s}{2} \log \pi + \log \Gamma(s/2).$$

The formula for $E'_{\ell,N}$ follows from $\frac{d}{ds} \ell^{-rs} = -(r \log \ell) \ell^{-rs}$. The basic phase derivative lemma completes the proof. $\square$

4 The archimedean chart and Riemann–Siegel theta

On the critical line, write

$$s = \frac{1}{2} + it.$$

The Riemann–Siegel theta function is

$$\vartheta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi.$$

It is the archimedean phase that makes

$$Z(t) = e^{i\vartheta(t)} \zeta(1/2 + it)$$

real for real t .

Proposition 4.1 (Archimedean phase velocity). *For real t away from branch choices,*

$$\vartheta'(t) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi.$$

Moreover, the phase velocity of $C(1/2 + it)$ differs from $\vartheta'(t)$ only by the constant branch contribution of $\frac{1}{2}s(s-1)$, hence has the same derivative on intervals avoiding branch cuts.

Proof. Differentiate the defining expression for ϑ :

$$\frac{d}{dt} \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) = \operatorname{Im} \left(\frac{i}{2} \psi\left(\frac{1}{4} + \frac{it}{2}\right) \right) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right).$$

The derivative of $-t \log \pi/2$ is $-\log \pi/2$. On the critical line, $s(s-1) = -(t^2+1/4)$ is real negative, so its argument is constant between branch jumps. $\square$

5 Phase destruction

A zero cannot be detected by phase alone. A phase condition can be satisfied while the amplitude is large. The completed finite atlas therefore suggests a two-condition test:

$$|\mathcal{X}_{\mathcal{P},N}(1/2 + it)| \text{ small}, \quad \arg \mathcal{X}_{\mathcal{P},N}(1/2 + it) \equiv \frac{\pi}{2} \pmod{\pi}$$

or an equivalent Hardy-real normalization.

Definition 5.1 (Finite phase-destruction candidate). *A height t is a finite $(\mathcal{P}, N)$ -candidate if it is a local minimum of $|\mathcal{X}_{\mathcal{P},N}(1/2 + it)|$ and the phase of $\mathcal{X}_{\mathcal{P},N}(1/2 + it)$ lies within a prescribed tolerance of a destructive phase class modulo π .*

Phase velocity separates into archimedean and prime Pandrosion contributions

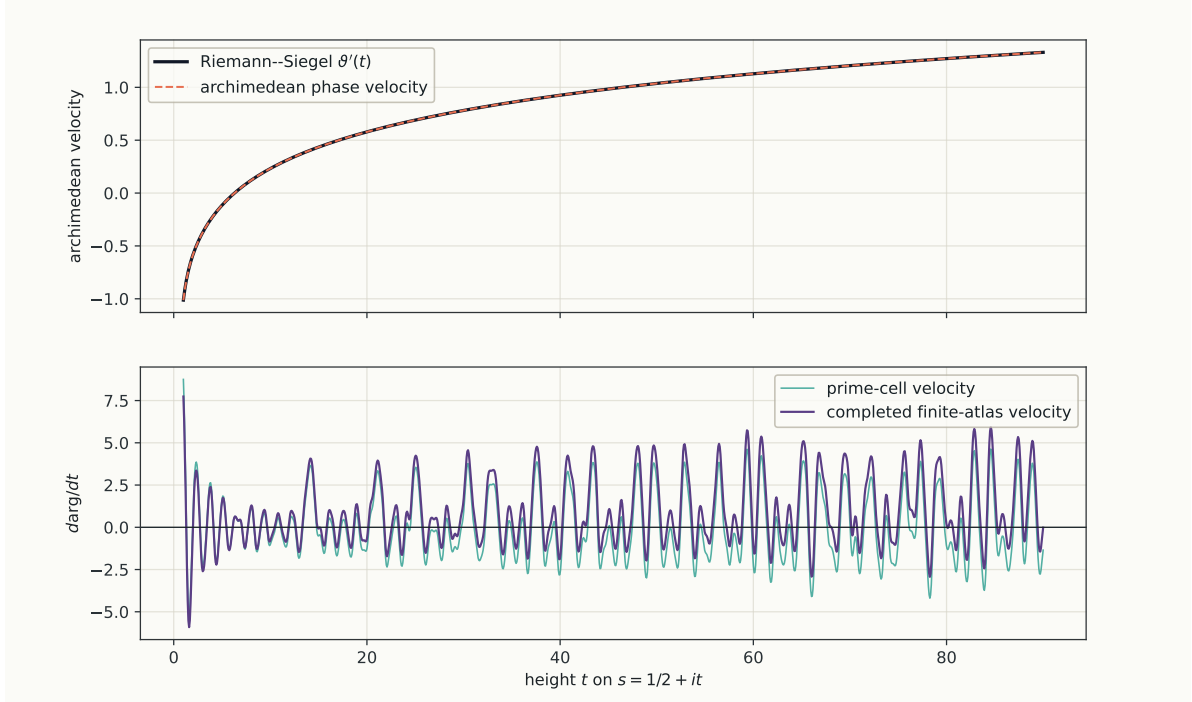


Figure 2: The completed phase velocity splits into an archimedean part and a finite prime-cell part. The upper panel verifies the Riemann–Siegel archimedean velocity; the lower panel adds the finite Euler-Pandrosion contribution.

Finite completed atlas: zero candidates require amplitude collapse and phase alignment

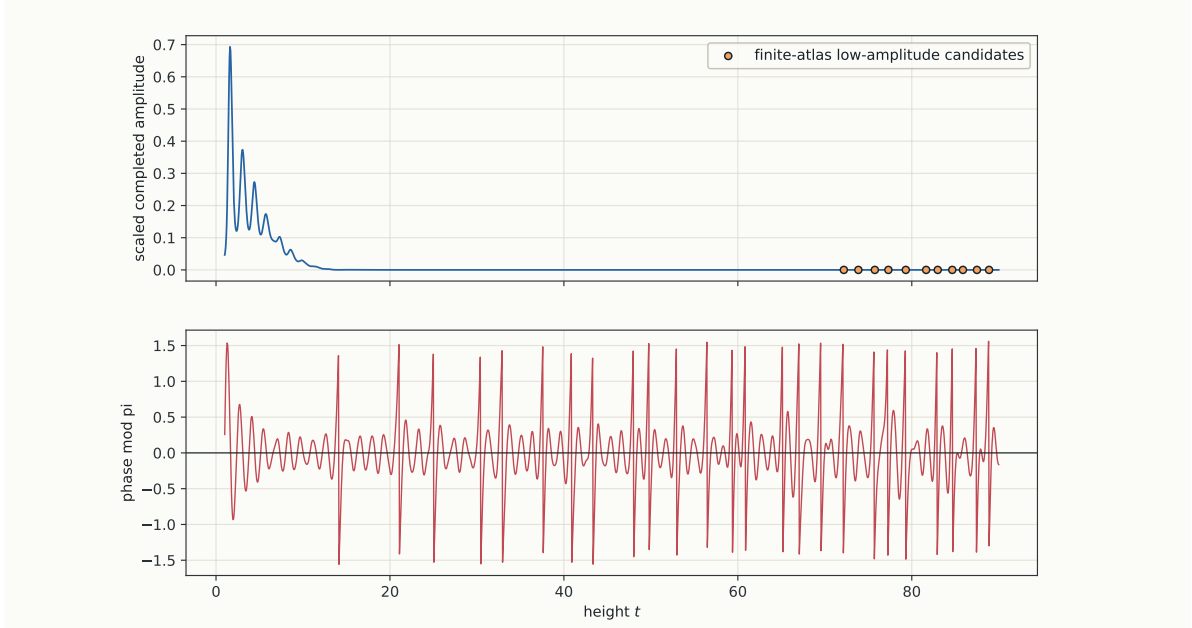


Figure 3: Finite completed atlas illustration. These plotted candidates are not zeta zeros; they show the intended two-filter logic: low completed amplitude plus phase alignment.

Conjecture 5.2 (Completed Pandrosion phase-destruction principle). *There is a renormalized exhaustion of prime sets $\mathcal{P}_L$, depths N_L , and archimedean normalizations such that the finite phase-destruction candidates of $\mathcal{X}_{\mathcal{P}_L, N_L}$ converge, with multiplicity, to the nontrivial zeros of the completed zeta function $\xi(s)$.*

Remark 5.3. *This conjecture is deliberately weaker than the Riemann hypothesis. It says that a Pandrosion phase model can approximate the zero set of ξ . A stronger version would assert that stable limiting candidates can occur only on the balanced line $\Re(s) = 1/2$, using the chart imbalance from Paper 025.*

6 What is new relative to 025

Paper 025 proves the finite direct/reciprocal balance theorem:

$$\mathcal{E}_+(\sigma) = \mathcal{E}_\infty(\sigma) \iff \sigma = \frac{1}{2}.$$

That explains why the critical line is geometrically selected by the prime-atlas charts. It does not supply the zero dynamics.

This paper adds:

1. the archimedean gamma chart $C(s)$;
2. the completed finite atlas $\mathcal{X}_{\mathcal{P}, N}$;
3. an exact phase-velocity formula;
4. the Riemann–Siegel theta derivative as the archimedean velocity;
5. a two-filter zero-candidate principle: amplitude collapse plus phase alignment.

7 Research targets

The phase calculus gives several concrete next tasks.

1. Define the correct renormalization of $\mathcal{X}_{\mathcal{P}, N}$ in the critical strip, since raw Euler products do not converge there.
2. Compare finite phase-destruction candidates with the Riemann–Siegel zero-counting formula.
3. Prove compactness for candidate sequences under prime-set/depth exhaustion.
4. Add the chart-imbalance energy from Paper 025 and prove that off-line candidates are unstable under completion.
5. Relate the completed Pandrosion phase velocity to the explicit formula, where primes and zeros already appear as dual spectral objects.

8 Conclusion

The most promising Pandrosion–zeta route is not a classical finite interpolation identity. It is a completed atlas: prime Euler factors as geometric Pandrosion cells, the gamma factor as the archimedean chart, and the critical line as the direct/reciprocal balance locus. Paper 025 supplied the balance theorem. This paper supplies the finite phase calculus. The resulting program is precise: construct a renormalized completed Pandrosion atlas whose phase-destruction candidates converge to the zeros of ξ , then use the chart imbalance to investigate why stable candidates should be confined to the critical line.

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